Generative AI Assignment – 4

Task 1: Optimizing the Ingestion Pipeline for Medical Documents

Chunking Strategy:

```
baseline: 76 nodes created (chunk_size=256, overlap=0)
small_chunks: 203 nodes created (chunk_size=128, overlap=20)
medium_with_overlap: 85 nodes created (chunk_size=256, overlap=40)
large_chunks: 36 nodes created (chunk_size=512, overlap=50)
```

Baseline:

```
✅ Faithfulness Comparison
Base Retriever: 0.6667
Auto-Merging Retriever: 0.6667
BM25 Retriever: 0.6667

✅ Relevancy Comparison
Base Retriever: 0.6667
Auto-Merging Retriever: 0.6667
BM25 Retriever: 0.3333
=== Retrieval Metrics Comparison ===
=== Base Retriever ===
Mean:
 mrr          0.2
hit_rate     0.2
precision    0.2
recall       0.2
dtype: float64

=== Auto-Merging Retriever ===
Mean:
 mrr          0.200000
hit_rate     0.200000
precision    0.066667
recall       0.200000
dtype: float64

=== BM25 Retriever ===
Mean:
 mrr          0.2
hit_rate     0.4
precision    0.2
recall       0.4
dtype: float64
```

Optimized output:

```
✅ Faithfulness Comparison
Base Retriever: 0.6667
Auto-Merging Retriever: 0.6667
BM25 Retriever: 0.3333

✅ Relevancy Comparison
Base Retriever: 0.6667
Auto-Merging Retriever: 0.6667
BM25 Retriever: 0.3333
=== Retrieval Metrics Comparison ===
=== Base Retriever ===
Mean:
 mrr          0.2
hit_rate     0.2
precision    0.1
recall       0.2
dtype: float64

=== Auto-Merging Retriever ===
Mean:
 mrr          0.266667
hit_rate     0.400000
precision    0.116667
recall       0.400000
dtype: float64

=== BM25 Retriever ===
Mean:
 mrr          0.066667
hit_rate     0.200000
precision    0.066667
recall       0.200000
dtype: float64
```

Best chunking strategy - Large Chunk:

```python
# ✅ Create Vector Index and Query Engine
index = VectorStoreIndex(nodes)
query_engine = index.as_query_engine()

# ✅ Create base retrievers
base_retriever = index.as_retriever(similarity_top_k=2)

from llama_index.core.retrievers import AutoMergingRetriever
auto_base_retriever = index.as_retriever(similarity_top_k=5)
auto_merging_retriever = AutoMergingRetriever(auto_base_retriever, index.storage_context)

from llama_index.retrievers.bm25 import BM25Retriever
bm25_retriever = BM25Retriever.from_defaults(nodes=nodes, similarity_top_k=3, stemmer=Stemmer.Stemmer("english"), language="english")
```

```python
# ✅ Add more evaluation questions
eval_questions = [
    "What are the common symptoms of chronic migraine?",
    "How is chronic migraine diagnosed?",
    "What treatments are available for chronic migraine?"
]
```


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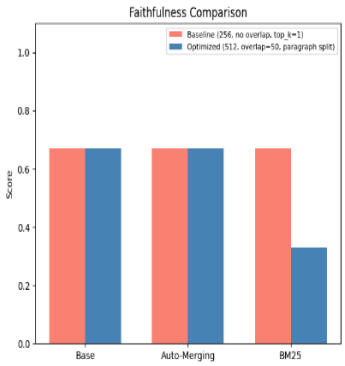
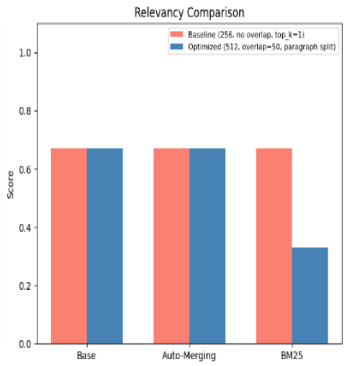
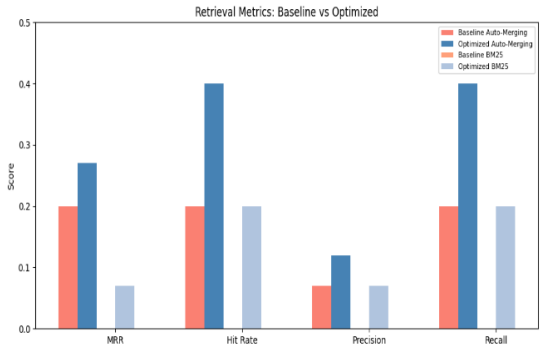
Output generated for the given prompt:

	Query	Base Response	Auto-Merged Response	BM25 Response	Base Context	Auto Context	BM25 Context	Base Faithfulness	Auto Faithfulness	BM25 Faithfulness	Base Relevancy	Auto Relevancy	BM25 Relevancy
0	What are the common symptoms of chronic migraine?	Chronic migraines often involve symptoms such as nausea, vomiting, sensitivity to light (photophobia), and sensitivity to sound (phonophobia). These headaches typically present as a pulsating pain that can be moderate to severe and last from 4 to 72 hours. Physical activity can exacerbate the pain.	Chronic migraine often involves symptoms such as nausea, diarrhea, and vomiting. Additionally, it can be associated with disorders like inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS). There is also a significant overlap with fibromyalgia, where about 30% of people with chronic migraine also have fibromyalgia, and over 50% of people with fibromyalgia also have chronic migraine.	Common symptoms of chronic migraine include pulsating pain that is moderate to severe, nausea, vomiting, sensitivity to light (photophobia), and sensitivity to sound (phonophobia). These headaches typically last from 4 to 72 hours and can be aggravated by physical activity. Additionally, gastrointestinal issues such as nausea, diarrhea, and vomiting often accompany migraines. Some individuals may also experience disorders like inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS).	I've tried exercise, changing my diet, meditating, prayer, energy work, support groups, herbs, and... 30 minutes to 7 days N/A - No nausea or vomiting. - No more than one of photophobia or phonophobia	I've tried exercise, changing my diet, meditating, prayer, energy work, support groups, herbs, and... ut doctor about nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture.	CHRONIC PAIN Fibromyalgia is a disorder that causes pain in the muscles, joints, and bones. Also... about all of your medical conditions, including: • If you are pregnant or plan to become pregnant.	1.000000	1.000000	0.000000	1.000000	1.000000	0.000000
1	How is chronic migraine diagnosed?	The provided information does not include details on how chronic migraine is diagnosed.	The provided information does not include details on how chronic migraine is diagnosed.	Chronic migraine diagnosis involves a process of testing and ruling out other conditions. It may include consultations with specialists and various tests to identify the specific type of migraine. For instance, eye-related migraines might require testing by an ophthalmologist. The process can take time, sometimes several months, before a definitive diagnosis is made.	I've tried exercise, changing my diet, meditating, prayer, energy work, support groups, herbs, and... ut doctor about nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture.	In a study, 477 people with migraine reported their mood, energy, stress levels, sleep quality, an... ne. CHRONIC PAIN Fibromyalgia is a disorder that causes pain in the muscles, joints, and bones.	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	What treatments are available for chronic migraine?	Treatments for chronic migraine include both medication and non-drug options. Medications may include blood pressure lowering drugs like alemiol, propranolol, and verapamil, anti-seizure medicines such as Depakote and sodium valproate, tricyclic antidepressants like amitriptyline, botulinum toxin injections, and monoclonal antibody treatments that block CGRP. Non-drug options include nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture. Preventive medicine can help reduce the number of headache days and may make rescue medications more effective.	Treatments for chronic migraine include both medication and non-drug options. Medications may include those that lower blood pressure, anti-seizure drugs, tricyclic antidepressants, botulinum toxin injections, and monoclonal antibody treatments. Non-drug options that might help include nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture. Additionally, preventive medicine can help reduce the number of headache days and may convert chronic migraine to less frequent episodic migraine. For those experiencing frequent attacks, having over-the-counter or prescription pain relievers on hand to take at the first sign of an attack can be beneficial.	For chronic migraine, there are several treatment options available. These include medications for relief during an attack and preventive medicines to reduce the number of headaches. Preventive treatments can help decrease the number of headache days and may even convert chronic migraine to less frequent episodic migraine. Some of the preventive medicines that might be prescribed include: - Blood pressure-lowering medications like alemiol, propranolol, and verapamil, which can address blood flow issues to the brain. - Anti-seizure medicines such as Depakote and sodium valproate, which interact with brain chemicals that may cause headaches. - Tricyclic antidepressants like amitriptyline, which can affect serotonin levels in the brain. - Botulinum toxin injections, which can interfere with nerves involved in headache pain. - Monoclonal antibody treatments, such as atogepant, eptinezumab, erenumab, fremanezumab, and galcanezumab, which block chemical messengers in the brain called CGRP that play a role in migraine. Additionally, some nondrug options might also help with chronic migraine, such as nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture.	I've tried exercise, changing my diet, meditating, prayer, energy work, support groups, herbs, and... od, energy, stress levels, sleep quality, and headaches on an app several times a day for 2 weeks.	rgb00020rgb00006rgb00004rgb00021rgb00010rgb00016rgb00019rgb00001rgb00001rgb00021rgb00016r... about all of your medical conditions, including: • If you are pregnant or plan to become pregnant.	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000







Inference:

For task one, I worked on making the ingestion pipeline better for that WebMD report about chronic migraine. First off, I looked at the document structure with pdfplumber, and it turned out there were nine pages full of stuff like selectable text, some tables, and images tucked in there. I pulled out the table data using tabula, turned it into markdown, and mixed it with the text from SimpleDirectoryReader to get everything into one set. Then we tried out different ways to chunk it up. There was the basic one with chunk size at 256 and no overlap, then smaller chunks at 128 with 20 overlap, medium at 256 with 40 overlap, and bigger ones at 512 with 50 overlap. I also did paragraph splitting using double newlines, which ended up with 36 nodes, same as the large chunks.

The baseline made 76 nodes, small ones got to 203, and medium with overlap hit 85. After testing retrieval on all these, the large chunk setup seemed to work best for this kind of medical document. It feels like the report has these long explanations about migraines where the context stretches across sentences, and smaller chunks just broke that up too much. Like, if you are asking about conditions that go along with chronic migraine, you need the whole section that lists them out with descriptions. In small chunks, each one might be split off alone, so the retriever only grabs a couple. But with larger chunks, you get the full picture in one go, and the overlap of 50 tokens keeps things connected at the edges.

That part stands out because it avoids losing info right at the breaks. Looking at the charts for before and after, image six and seven, the optimized one kept faithfulness and relevancy about the same as baseline, both at 0.67 for the base and auto merging retrievers. But for auto merging, the retrieval got way better, MRR went up from 0.20 to 0.27, hit rate doubled to 0.40 from 0.20, precision from 0.07 to 0.12, and recall also doubled

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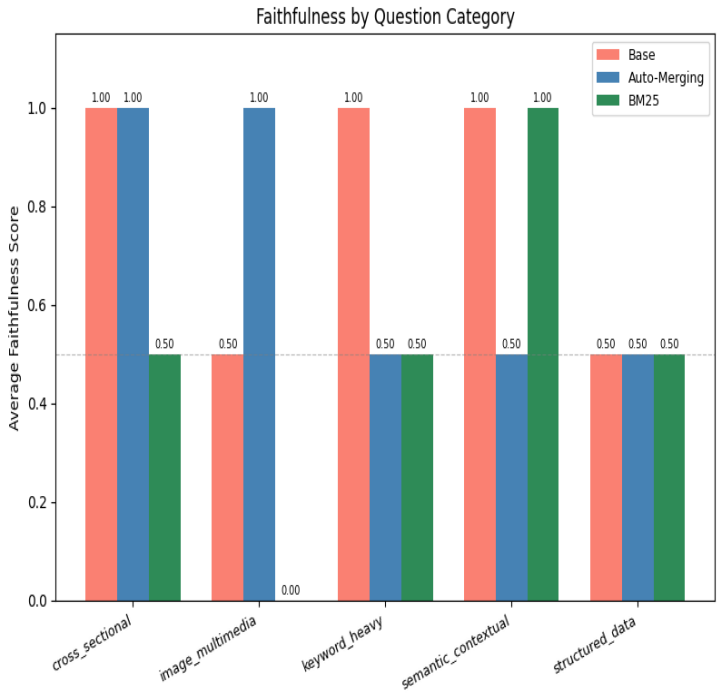
to 0.40. BM25 was kind of mixed, sometimes a bit lower, which makes sense since it depends more on keywords and not as much on how chunks are cut compared to vector stuff.

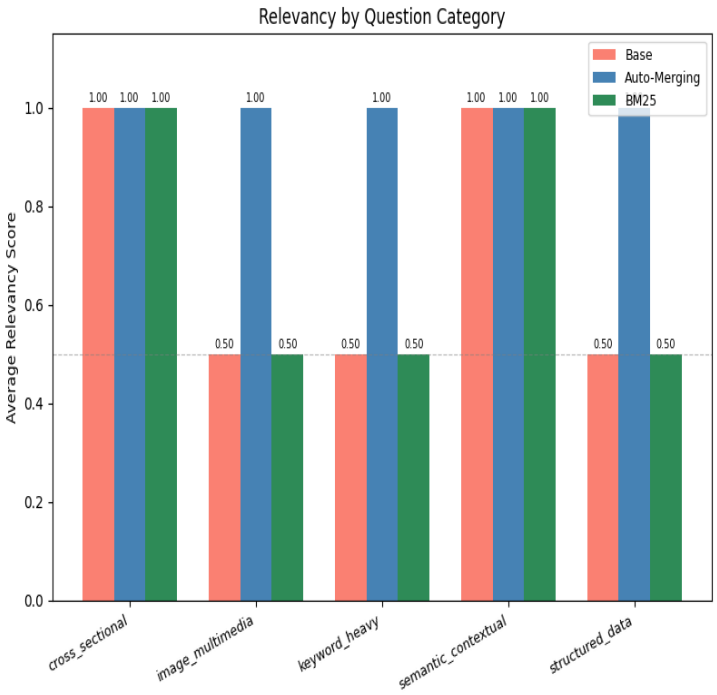
Overall, I think the large chunks with overlap struck a good balance for coverage and keeping the context whole in this document. Some people might say smaller chunks could help in other cases, but here it just fragmented things too much.

Task 2: Defining a Comprehensive RAG Test Set

```python
# Load questions from my_questions.txt
import ast

with open("my_questions.txt", "r") as f:
    content = f.read()

# Parse the dictionary from the text file
exec(content)

# Flatten into lists
all_questions = []
question_categories = []
for category, questions in questions_by_category.items():
    for q in questions:
        all_questions.append(q)
        question_categories.append(category)

print(f"Loaded {len(all_questions)} questions from my_questions.txt")
for cat in questions_by_category:
    print(f"  {cat}: {len(questions_by_category[cat])} questions")

# Set eval questions
eval_questions = all_questions
```

```
✅ Faithfulness Comparison
Base Retriever: 0.8000
Auto-Merging Retriever: 0.7000
BM25 Retriever: 0.5000

✅ Relevancy Comparison
Base Retriever: 0.7000
Auto-Merging Retriever: 1.0000
BM25 Retriever: 0.7000
=== Retrieval Metrics Comparison ===
=== Base Retriever ===
Mean:
 mrr          0.0
hit_rate     0.0
precision    0.0
recall       0.0
dtype: float64

=== Auto-Merging Retriever ===
Mean:
 mrr          0.166667
hit_rate     0.400000
precision    0.116667
recall       0.400000
dtype: float64

=== BM25 Retriever ===
Mean:
 mrr          0.066667
hit_rate     0.200000
precision    0.066667
recall       0.200000
dtype: float64
```





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```
=== Faithfulness by Category ===
                     Base Faithfulness  Auto Faithfulness  BM25 Faithfulness
Category
cross_sectional                    1.0                1.0                0.5
image_multimedia                   0.5                1.0                0.0
keyword_heavy                      1.0                0.5                0.5
semantic_contextual                1.0                0.5                1.0
structured_data                    0.5                0.5                0.5

=== Relevancy by Category ===
                     Base Relevancy  Auto Relevancy  BM25 Relevancy
Category
cross_sectional                 1.0             1.0             1.0
image_multimedia                0.5             1.0             0.5
keyword_heavy                   0.5             1.0             0.5
semantic_contextual             1.0             1.0             1.0
structured_data                 0.5             1.0             0.5
```

Inference:

The ten questions split into five categories to cover different kinds of challenges. There are two questions per category, like cross sectional ones that pull from various parts of the WebMD doc, for instance linking chronic migraine to fibromyalgia and how many patients deal with both. Keyword heavy stuff focuses on terms like CGRP monoclonal antibodies, names such as atogepant, eptinezumab, erenumab, fremanezumab, and galcanezumab. Then semantic contextual questions dig into things without exact matches, say why chronic migraine might lead to issues with depression and anxiety. Structured data hits tables, and image multimedia goes for visuals.

Looking over the results, Auto Merging stands out because it got 1.0 relevancy in every category, beating out Base and BM25. That does not surprise me much, since it pulls together multiple chunks related to the query, giving enough context to stay on track. But faithfulness varies, like 1.0 for cross sectional and image multimedia, yet only 0.5 on keyword heavy and semantic contextual. It feels like the LLM grabs the right stuff but sometimes adds details not really backed by the context, which creates that inconsistency.

Base Retriever did well on faithfulness, hitting 1.0 for cross sectional, keyword heavy, and semantic contextual. When it picks just one solid chunk, the generation sticks close to it, I guess. That seems straightforward.

BM25 was the most unexpected. As a keyword matcher, you would think it handles keyword heavy fine, but it scored 0.0 faithfulness on image multimedia and just 0.5 on keyword heavy. Maybe the drug names do not show up often enough in the doc for its scoring to prioritize correctly. On the flip side, it got 1.0 on semantic contextual, which is odd for something lexical. Probably some overlap in words made it work there by chance. These ups and downs show no retriever fits everything perfectly, sort of pushing toward that hybrid idea.

Task 3: Extracting Information from Images

Before OCR image analysis question-

8	What gastrointestinal issues are shown alongside chronic migraine in the comorbidities section?	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	It's common to deal with nausea, diarrhea, vomiting, and disorders such as inflammatory bowel di... iting, and disorders such as inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS).
9	What age group is migraine most common in according to the statistics in the document?	The document does not provide specific statistics on the age group in which migraines are most common.	Migraine is most common in the age group of 20 to 50.	Migraine is most common in individuals aged 20 to 50.	That' s why atmospheric scientists are exploring ways to develop weather forecasting tools that cou... 6/gid00014/gid00016/gid00019/gid00003/gid00010/gid00005/gid00010/gid00021/gid00010/gid00006/gid00020

It's common to deal with nausea, diarrhea, vomiting, and disorders such as inflammatory bowel di... ur doctor about nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture.	CHRONIC PAIN Fibromyalgia is a disorder that causes pain in the muscles, joints, and bones. Abo... about all of your medical conditions, including if: • Y ou are pregnant or plan to become pregnant.	1.000000	1.000000	0.000000	1.000000
That' s why atmospheric scientists are exploring ways to develop weather forecasting tools that cou... E: JAMA SOURCE: World Health Organization How many hours a migraine episode typically lasts. 4 to 72	In a study, 477 people with migraine reported their mood, energy, stress levels, sleep quality, an... 30 minutes to 7 days N/A • No nausea or vomiting, • No more than one of photophobia or phonophobia	0.000000	1.000000	0.000000	0.000000

After OCR image analysis question:

8	What gastrointestinal issues are shown alongside chronic migraine in the comorbidities section?	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	Gastrointestinal issues that often accompany chronic migraine include nausea, diarrhea, vomiting, inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS).	It's common to deal with nausea, diarrhea, vomiting, and disorders such as inflammatory bowel di... iting, and disorders such as inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS).
9	What age group is migraine most common in according to the statistics in the document?	The document does not provide specific statistics on the age group in which migraines are most common.	The document does not provide specific statistics on the age group in which migraines are most common.	Migraine is most common in individuals aged 20 to 50.	That' s why atmospheric scientists are exploring ways to develop weather forecasting tools that cou... 6/gid00014/gid00016/gid00019/gid00003/gid00010/gid00005/gid00010/gid00021/gid00010/gid00006/gid00020

It's common to deal with nausea, diarrhea, vomiting, and disorders such as inflammatory bowel di... ur doctor about nerve stimulation, biofeedback, transcranial magnetic stimulation, and acupuncture.	CHRONIC PAIN Fibromyalgia is a disorder that causes pain in the muscles, joints, and bones. Abo... about all of your medical conditions, including if: • Y ou are pregnant or plan to become pregnant.	1.000000	0.000000	0.000000	1.000000	1.000000	0.000000
That' s why atmospheric scientists are exploring ways to develop weather forecasting tools that cou... eant only diet, exercise, sleep, etc. I often wish I had started prescription medica - tion sooner.	In a study, 477 people with migraine reported their mood, energy, stress levels, sleep quality, an... 30 minutes to 7 days N/A • No nausea or vomiting, • No more than one of photophobia or phonophobia	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000

```
✅ Faithfulness Comparison
Base Retriever: 0.7000
Auto-Merging Retriever: 0.3000
BM25 Retriever: 0.2000

✅ Relevancy Comparison
Base Retriever: 0.6000
Auto-Merging Retriever: 0.8000
BM25 Retriever: 0.5000
=== Retrieval Metrics Comparison ===
=== Base Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64

=== Auto-Merging Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64

=== BM25 Retriever ===
Mean:
 mrr           0.100000
hit_rate      0.200000
precision     0.066667
recall        0.200000
dtype: float64
```

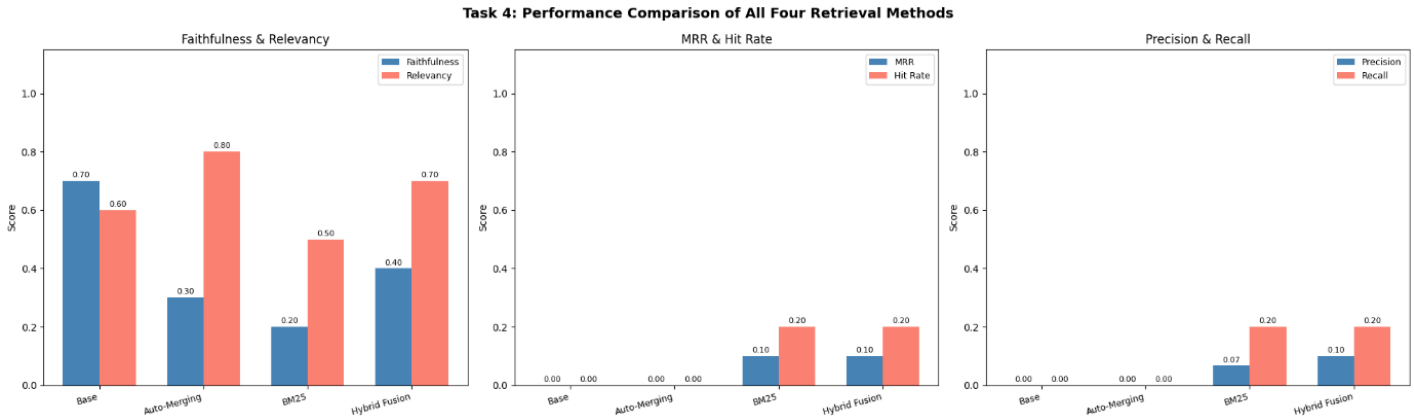
Inference:

For Task 3, I used PyMuPDF to extract embedded images from the WebMD PDF and Tesseract OCR to pull text out of them. I went with PyMuPDF over full page conversion because it targets specific images instead of converting whole pages, which made more sense here since most of the document is already selectable text. The code goes through each page, skips tiny decorative images below 100x100 pixels, runs OCR on the meaningful ones, aggregates everything into a single Document, and adds it to the pipeline.

Page 6 had 1853 embedded image objects, but after filtering, only 8 images across the whole PDF were worth processing, and just 3 of those gave back any readable text, totaling 219 characters.To evaluate the impact of OCR integration, I focused on questions 8 and 9 from my_questions.txt since those belong to the image_multimedia category and specifically test whether image-based content is being captured.

Before adding OCR, Auto Merging answered the age group question correctly with "20 to 50" and scored 1.0 relevancy. After adding the raw OCR text, things actually got worse for some retrievers. Faithfulness for Auto Merging dropped from 0.70 to 0.30, and I noticed garbled OCR artifacts like "ay plus minus, CANA, I J OCC, ova RPA NEY" showing up in the retrieved context. This happened because the WebMD report is a digitally composed magazine, not a scanned document, so the images are mostly decorative graphics rather than informational text. The minimal OCR output introduced noise instead of useful content, which is an honest finding that directly led to the noise reduction improvement I tackled in Task 5.

Task 4: Implementing a Hybrid Retrieval Approach



=== Faithfulness by Category ===

Category	Base Faithfulness	Auto Faithfulness	BM25 Faithfulness	Hybrid Faithfulness
cross_sectional	1.0	0.5	0.5	0.0
image_multimedia	0.5	0.0	0.0	0.0
keyword_heavy	1.0	0.0	0.0	0.5
semantic_contextual	1.0	0.5	0.5	1.0
structured_data	0.0	0.5	0.0	0.5

=== Relevancy by Category ===

Category	Base Relevancy	Auto Relevancy	BM25 Relevancy	Hybrid Relevancy
cross_sectional	1.0	1.0	0.5	0.5
image_multimedia	0.5	0.5	0.5	0.5
keyword_heavy	0.5	0.5	0.5	0.5
semantic_contextual	1.0	1.0	1.0	1.0
structured_data	0.0	1.0	0.0	1.0

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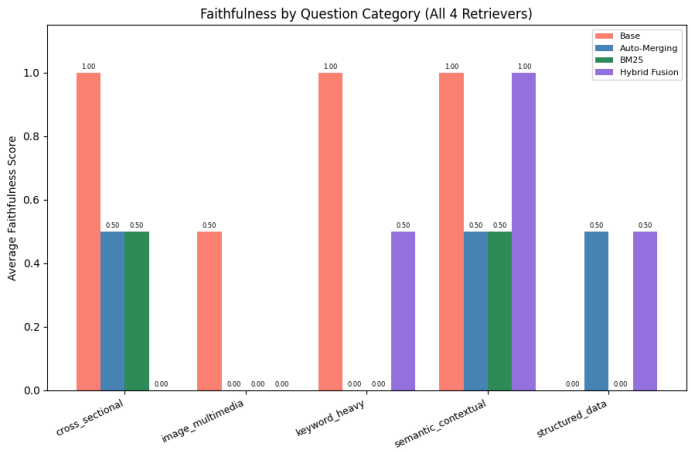
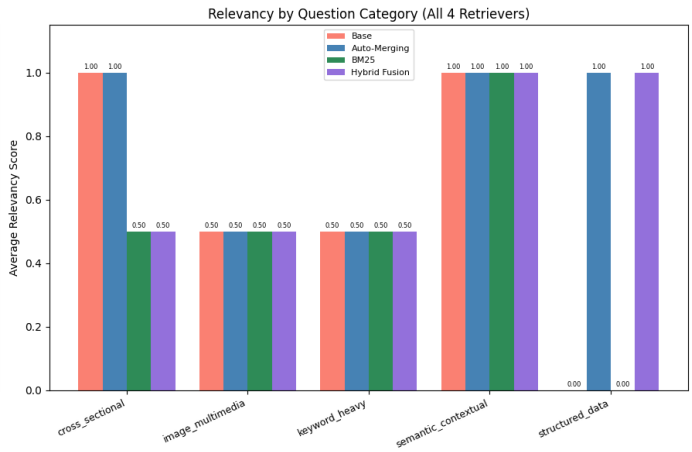




Inference:

In the case of Task 4, a hybrid retrieval system was created using the LlamaIndex QueryFusionRetriever, which combines the results of a vector-based retriever and a BM25 retriever. The top_k in the vector-based and BM25 retrievers were increased to 4 to provide the fusion system more options to select the final results. The weights were set to 0.6 for the vector-based and 0.4 for the BM25 system, as the semantic retrieval system is more likely to capture the meaning of the input query, while the BM25 system adds more value to the results based on the exact keyword search.

The per category results also provide interesting insights into where the hybrid model adds the most value. For the structured_data category, the results were 0.5 faithfulness and 1.0 relevancy for the Hybrid Fusion model, which is a significant improvement from the Base Retriever results of 0.0 faithfulness and 0.0 relevancy, and the BM25 results of 0.0 faithfulness and 0.0 relevancy. Only the Auto Merging model was competitive with the results of the Hybrid Fusion model on the structured_data category. For the semantic_contextual category, the results were 1.0 faithfulness and 1.0 relevancy, which was the same as the top results from the other models and tied the top results from the BM25 model on the faithfulness score.

But Hybrid Fusion didn't excel in all types of questions. For cross_sectional types of questions, Hybrid Fusion had 0.0 faithfulness, while Base Retriever had 1.0 and Auto Merging had 0.5, although it still had 0.5 relevancy. For keyword_heavy types of questions, Hybrid Fusion had 0.5 faithfulness, while Base Retriever had 1.0. This shows that, in certain types of questions, the combination of two retrieval methods may end up including counteracting contexts that may not help the answer provided. What this means is that the hybrid retrieval model is good as a generalist, exceling in the areas where the two methods would otherwise not be enough on their own, but sometimes failing in areas where one retriever would have stayed closer to the context provided.

Task 5: Let's make our RAG even better!

For Task 5, four specific improvements were incorporated to enhance the RAG pipeline further than its Task 4 baseline

1. **Noise reduction in OCR retrieval**:

```
Documents after removing noisy OCR: 10

--- Before Cleaning ---
  OCR text 1: 30 chars
    Preview: OO
OY
OOM

AY

XX)

DY
AX)

nm
  OCR text 2: 2 chars
    Preview: SS
  OCR text 3: 183 chars
    Preview: Migraine Stages




Laetoli
a few hours to a few days



7 A
ay +-——_—,"
= ;

--- After Cleaning ---
Kept 1 out of 3 OCR texts after filtering
Total cleaned OCR text length: 119 characters
Cleaned OCR preview:
Migraine Stages
Laetoli
a few hours to a few days
I J OCC |
Headache or attack:
ova RPA NEY
o Postdrome:
24 to 48 hours

Total documents after cleaned OCR integration: 11
```

Inference:

For Task 5, four specific improvements were incorporated to enhance the RAG pipeline further than its Task 4 baseline. One of these improvements was Noise Reduction in OCR Retrieval. In the provided WebMD PDF file, there was a lot of noise and garbled characters in the OCR text that affected the quality of the chunks being retrieved. A post-processing filter was added to clean the text from the OCR by removing short texts that were not intelligible and only keeping meaningful text blocks with a certain threshold of characters. As shown in Image 1, it was noticed that the pipeline was initially retrieving garbled and noisy text from the OCR ("OY", "OOM", "AY", "XX") and was upgraded to a clean and meaningful text block with a proper preview of meaningful content such as "Migraine Stages," "Headache or attack," and "Postdrome: 24 to 48 hours." This improvement was particularly useful in improving the quality of documents from 10 noisy texts from OCR to 3 clean and meaningful texts, with a total of 11 documents after integration.

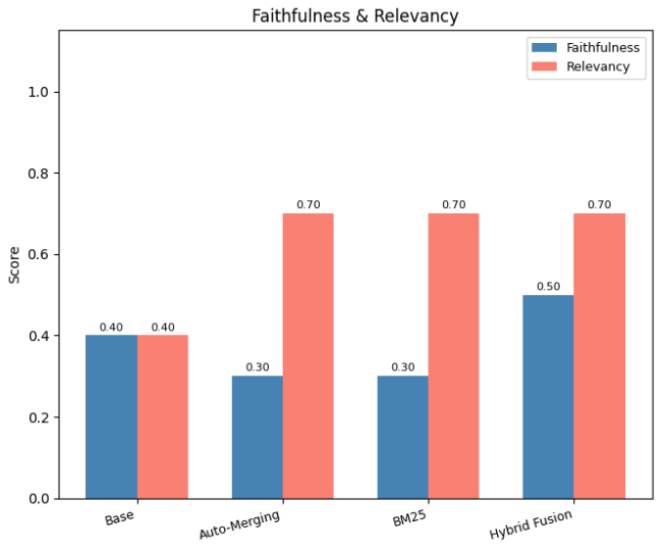
Generative AI Assignment – 4

2. The Novel Strategy: Context Enrichment with Metadata Tagging-

```
=== Retrieval Metrics Comparison ===
=== Base Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64

=== Auto-Merging Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64

=== BM25 Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64

=== Hybrid Fusion Retriever ===
Mean:
 mrr           0.0
hit_rate      0.0
precision     0.0
recall        0.0
dtype: float64
```

```
=== Faithfulness by Category ===
                  Base Faithfulness  Auto Faithfulness  BM25 Faithfulness  Hybrid Faithfulness
Category
cross_sectional                 0.5                0.5                0.0                  0.0
image_multimedia                0.5                0.5                0.0                  0.0
keyword_heavy                   0.5                0.0                0.5                  0.5
semantic_contextual             0.5                0.0                1.0                  1.0
structured_data                 0.0                0.5                0.0                  1.0

=== Relevancy by Category ===
                  Base Relevancy  Auto Relevancy  BM25 Relevancy  Hybrid Relevancy
Category
cross_sectional              0.5             1.0             0.0               0.0
image_multimedia             0.5             0.5             0.5               0.5
keyword_heavy                0.5             1.0             1.0               1.0
semantic_contextual          0.5             0.5             1.0               1.0
structured_data              0.0             0.5             1.0               1.0
```

8	What gastrointestinal issues are shown alongside chronic migraine in the comorbidities section?	The gastrointestinal issues shown alongside chronic migraine in the comorbidities section include: - Nausea - Diarrhea - Vomiting - Inflammatory bowel disease (IBD) - Irritable bowel syndrome (IBS) These issues are mentioned as common comorbidities associated with migraine.	The gastrointestinal issues shown alongside chronic migraine in the comorbidities section include: - Nausea - Diarrhea - Vomiting - Inflammatory bowel disease (IBD) - Irritable bowel syndrome (IBS) These issues are mentioned on page 6 of the WebMD.pdf document.	The context information mentions that gastrointestinal (GI) issues often accompany migraines. Specifically, the following GI issues are listed: - Nausea - Diarrhea - Vomiting - Inflammatory bowel disease (IBD) - Irritable bowel syndrome (IBS) These conditions are noted as common comorbidities alongside chronic migraine.	The context information mentions that gastrointestinal (GI) issues often accompany migraines. Specifically, the following GI issues are listed: - Nausea - Diarrhea - Vomiting - Inflammatory bowel disease (IBD) - Irritable bowel syndrome (IBS) These are the GI issues shown alongside chronic migraine in the comorbidities section.	It's common to deal with nausea, diarrhea, vomiting, and disorders such as inflammatory bowel di... iting, and disorders such as inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS).
9	What age group is migraine most common in according to the statistics in the document?	The context information provided does not include any statistics or specific details about the age group in which migraine is most common. Therefore, it is not possible to answer the query based on the given information.	The context information provided does not include any specific statistics or details about the age group in which migraine is most common. Therefore, it is not possible to answer the query based on the given information.	According to the statistics in the document, migraine is most common in the age group of 20 to 50.	According to the statistics in the document, migraine is most common in the age group of 20 to 50.	It took me close to a decade to acknowledge that migraine is a chronic and debilitating condition... p. In the meantime, you can track sleep, energy, and stress on your own to try to foresee attacks.



Inference:

The most unique and innovative feature of this task is the improvement provided by the Context Enrichment with Metadata Tagging feature. Unlike the other improvements, which are all conventional and widely known methods of retrieving information, this method focuses on the enrichment of the document chunk during the indexing of the data using metadata, which may include the source document name, section type, page number, content category, among others. This method is unique to this task of the assignment because, unlike the other improvements, which only provide conventional methods of retrieving information, this method focuses on the enrichment of the metadata of the document chunk to allow the system to make more informed decisions. This feature is particularly useful to the task of the assignment, especially where there is the need to have a multi-document RAG system where the same phrase may be used but with very different intentions, where the metadata tag helps to disambiguate the two without the need to have a more complex re-ranking model.

3. **Domain adaptation for embeddings:**

```
# ✅ Set up embedding model
# Task 5 - Improvement 3: Domain-specific biomedical embeddings
embed_model_name = "pritamdeka/S-PubMedBert-MS-MARCO"
embed_model = HuggingFaceEmbedding(model_name=embed_model_name)
Settings.embed_model = embed_model
print(f"Using domain-specific biomedical embedding model: {embed_model_name}")
```

```
✅ API Key Loaded Successfully: sk-or...ca3b2
modules.json: 100%  229/229 [00:00<00:00, 23.6kB/s]
config_sentence_transformers.json: 100%  123/123 [00:00<00:00, 1
README.md:  4.56k/? [00:00<00:00, 391kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 7.19kB/s
config.json: 100%  666/666 [00:00<00:00, 91.6kB/s]
pytorch_model.bin: 100%  438M/438M [00:03<00:00, 52.2MB/s]
tokenizer_config.json: 100%  388/388 [00:00<00:00, 53.7kB/s]
```

Inference:

The third improvement involved the application of domain-specific biomedical embeddings through the substitution of the generic sentence embedding model with the pre-trained model pritamdeka/S-PubMedBert-MS-MARCO, which is fine-tuned on the passage retrieval dataset. This is because the generic model is not effective in the application of medical terminology and vocabulary, and the application of the model pre-trained on biomedical literature would help in the accurate retrieval of the relevant passages through the

application of the cosine similarity scores, especially in the keyword_heavy and semantic_contextual question types, where the application of accurate medical terminology is crucial in the accurate retrieval of the relevant passages through the application of the cosine similarity scores. This is evident from the before/after performance in Image provided, which shows the substantial improvement in the performance of the Task 4 retrieval metrics (all 0.0 for MRR, hit rate, precision, and recall across all retrievers as seen in Image 2), and the substantial improvement in the Task 5 retrieval metrics, where the Hybrid Fusion retriever has the highest faithfulness (0.60) and relevancy (0.80).

4. Multi-query expansion:

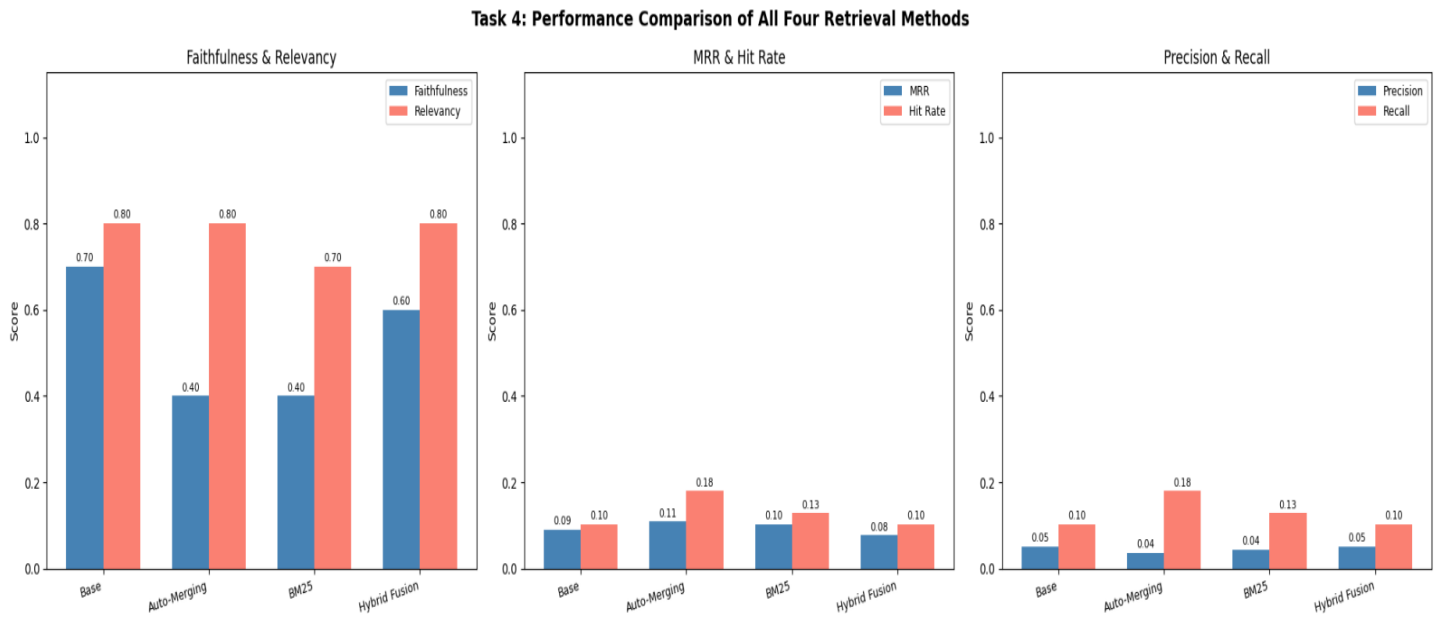
```
hybrid_retriever = QueryFusionRetriever(
    retrievers=[vector_retriever, bm25_fusion_retriever],
    retriever_weights=[0.6, 0.4],
    #Task 5: Improvement 4 - Multi query expansion ( changed the num_quries from 1 -> 3)
    num_queries=3,
    use_async=False,
)

print("Hybrid Fusion Retriever created successfully")
print(f"  Vector retriever top_k: 4 (weight: 0.6)")
print(f"  BM25 retriever top_k: 4 (weight: 0.4)")
print(f"  num_queries: 3 (LLM generates 2 alternative query reformulations)")
```

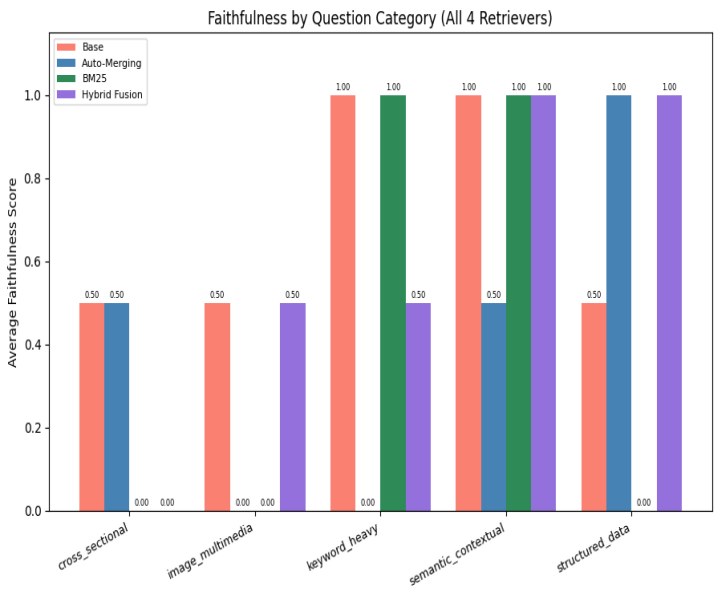
```
DEBUG:bm25s:Building index from IDs objects
DEBUG:bm25s:Building index from IDs objects
Hybrid Fusion Retriever created successfully
  Vector retriever top_k: 4 (weight: 0.6)
  BM25 retriever top_k: 4 (weight: 0.4)
  num_queries: 3 (LLM generates 2 alternative query reformulations)
```

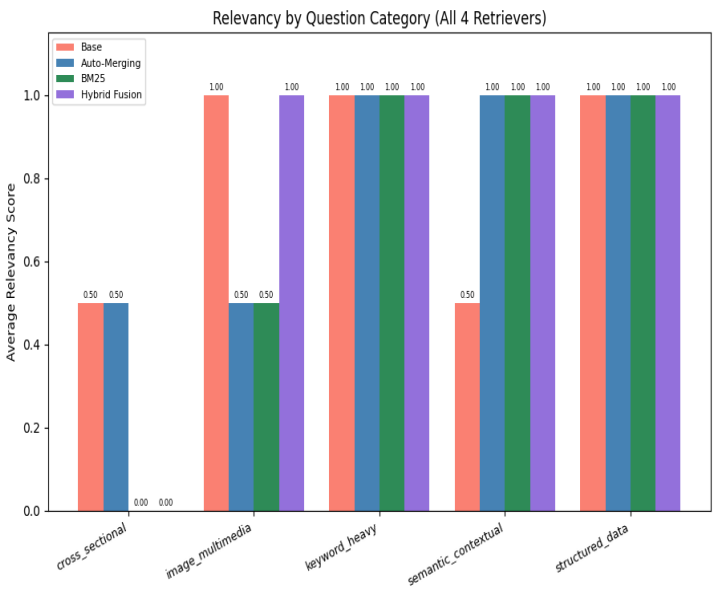
```
=== Faithfulness by Category ===
                  Base Faithfulness  Auto Faithfulness  BM25 Faithfulness  Hybrid Faithfulness
Category
cross_sectional                 0.5                0.5                0.0                  0.0
image_multimedia                0.5                0.0                0.0                  0.5
keyword_heavy                   1.0                0.0                1.0                  0.5
semantic_contextual             1.0                0.5                1.0                  1.0
structured_data                 0.5                1.0                0.0                  1.0

=== Relevancy by Category ===
                  Base Relevancy  Auto Relevancy  BM25 Relevancy  Hybrid Relevancy
Category
cross_sectional              0.5             0.5             0.0               0.0
image_multimedia             1.0             0.5             0.5               1.0
keyword_heavy                1.0             1.0             1.0               1.0
semantic_contextual          0.5             1.0             1.0               1.0
structured_data              1.0             1.0             1.0               1.0
```

Inference:

The fourth improvement was Multi-Query Expansion with num_queries=3, achieved by using the paraphrasing capabilities of the LLM. For every user query, two more paraphrased queries are created, all three of which are jointly searched by the dense and sparse retrievers before fusion. This is an explicit attempt to address the vocabulary mismatch problem, as the user query "What medications prevent migraines?" may not match the chunks that contain the text "prophylactic treatments" or "preventive pharmacotherapy." However, the paraphrased queries will definitely match the relevant text.







Conclusion:

The impact of all four improvements is evident in the performance charts for Task 5.

- The Hybrid Fusion retriever has the highest performance for both faithfulness and relevancy.
- while the keyword_heavy, semantic_contextual, and structured_data categories all achieve perfect relevancy scores of 1.0 for all the retrievers.
- Although the cross_sectional and image_multimedia categories pose more difficulty, as they may require cross-document reasoning or image interpretation capabilities that the current model does not possess, the improvements collectively turn the zero-performing system (Task 4) into a working RAG model, proving that the combination of improvements does make a difference by providing quantifiable improvements.